# TrustLens

A reliability score for every number on a dashboard, with the reason it dropped.

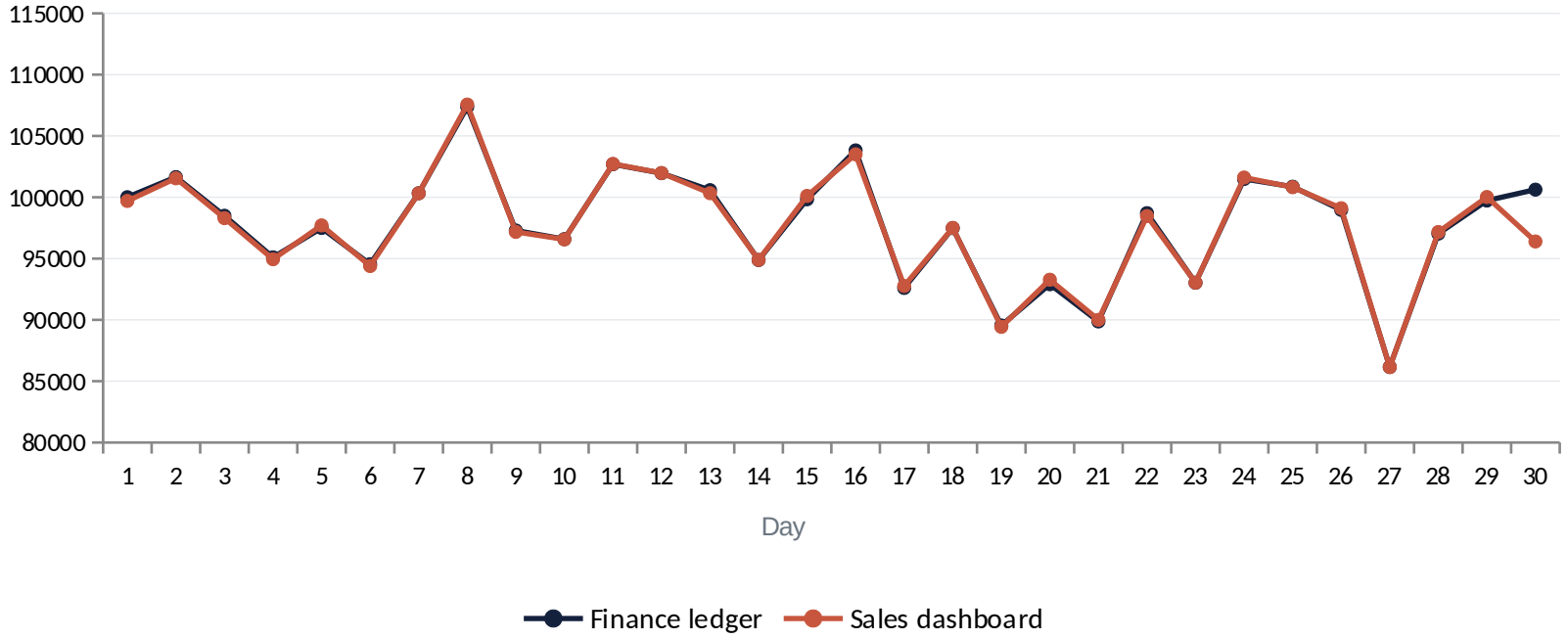
Hackathon Track 04: The Work Behind the Work
Problem: dashboards the business no longer trusts

# Same day, same metric, and the numbers still disagree

| Source | Revenue, day 30 | Note |
|---|---|---|
| Finance ledger | ₹1,00,623 | Source of truth, posted nightly |
| Sales dashboard | **₹96,397** | Built on orders table, loaded 9 hours late |
| Gap | ₹4,226 | -4.2% against the ledger |

Nothing on the dashboard says it is incomplete. A person reading it has no way to know, so the number gets quoted in a meeting.

# Thirty days of data: it only breaks on the last day



# Three checks scored the broken day. One of them saw nothing.

| Check | What it measures | Result | Score |
|---|---|---|---|
| Freshness | Hours past the expected load time | 9 h late | 52 |
| Consistency | Gap to the finance ledger | -4.2% | 50 |
| Anomaly | Z-score vs previous 14 days | z = 0.1 (normal) | 100 |

## Trust score: 65 / 100

Normal days averaged 99. Weights used: freshness 30%, consistency 40%, anomaly 30%.

# The score comes with a reason a manager can read

> Revenue (day 30) is low-confidence, score 65.
> The orders table loaded 9 hours after its usual time, and the total is 4.2% below the finance ledger (₹4,226). Normal day-to-day movement does not explain it, so treat this as a data problem, not a sales drop.

**Lineage: orders_raw → orders_clean → revenue_daily → Revenue tile**

The check engine marks orders_clean as the first stage that failed its freshness SLA.

# What is built and what comes next

## Prototype

- Python and DuckDB check engine
- Freshness, reconciliation and z-score checks
- Weighted trust score per metric
- Streamlit view with explanations from the Claude API

## Next

- Connect to a real warehouse and BI tool
- Learn per-metric thresholds from history
- Alert owners when a score drops below target
- Automatic lineage from SQL and job logs